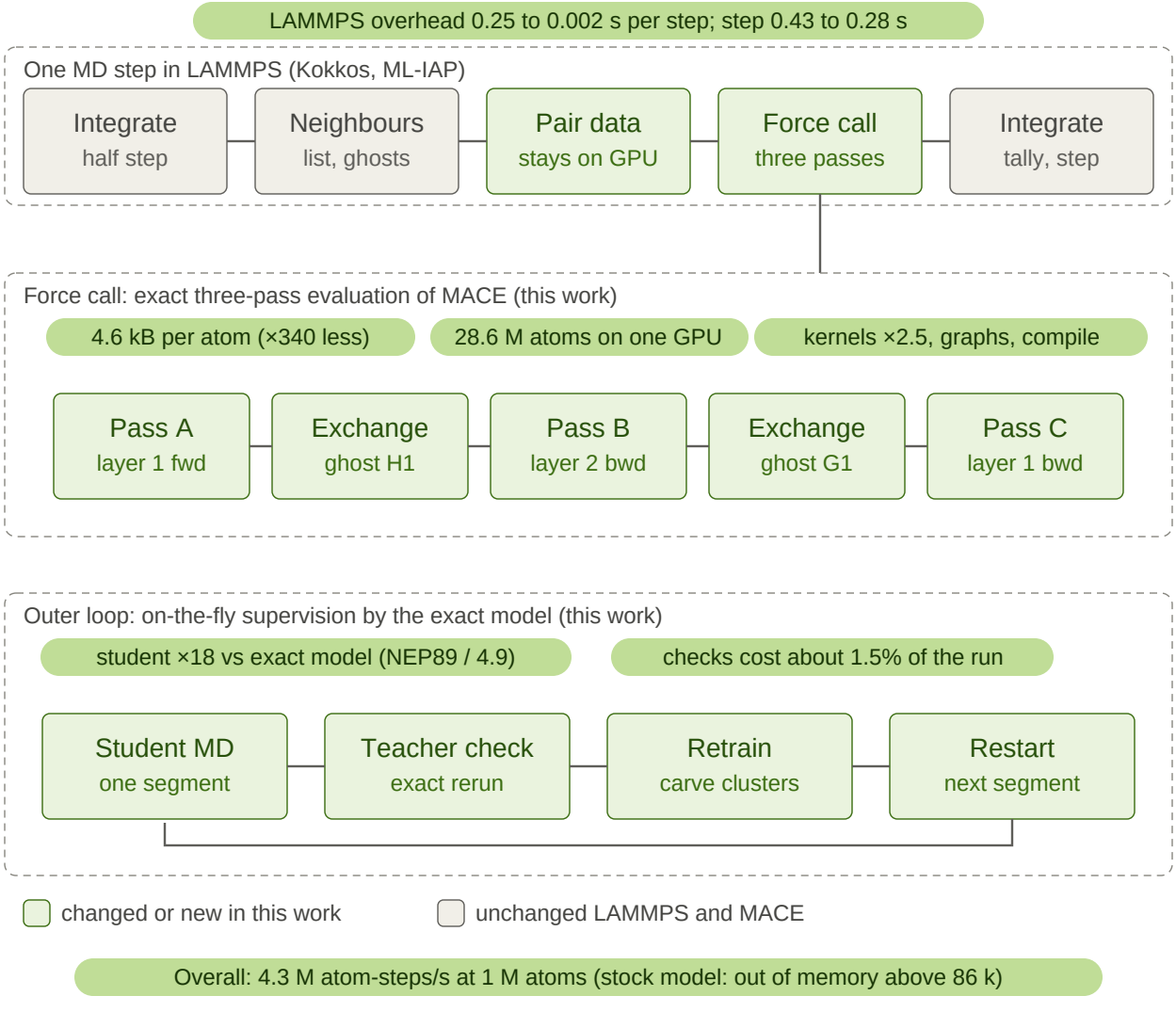
LAMMPS overhead 0.25 to 0.002 s per step; step 0.43 to 0.28 s
One MD step in LAMMPS (Kokkos, ML-IAP)
Integrate
half step
Neighbours
list, ghosts
Pair data
stays on GPU
Force call
three passes
Integrate
tally, step
Force call: exact three-pass evaluation of MACE (this work)
4.6 kB per atom (×340 less)
28.6 M atoms on one GPU
kernels ×2.5, graphs, compile
Pass A
layer 1 fwd
Exchange
ghost H1
Pass B
layer 2 bwd
Exchange
ghost G1
Pass C
layer 1 bwd
Outer loop: on-the-fly supervision by the exact model (this work)
student ×18 vs exact model (NEP89 / 4.9)
checks cost about 1.5% of the run
Student MD
one segment
Teacher check
exact rerun
Retrain
carve clusters
Restart
next segment
changed or new in this work
unchanged LAMMPS and MACE
Overall: 4.3 M atom-steps/s at 1 M atoms (stock model: out of memory above 86 k)